BC Wildlife-Aware Dashcam

Project Design — Proof-of-Concept Sprint, All Features Retained

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A small computer mounts on a car windshield and watches the road. When an animal is ahead, it recognizes what kind it is, estimates how close it is, and warns the driver — a beep and a light that grow more urgent the nearer the animal. Every sighting is saved with GPS location and time, building the first species-level map of where wildlife crosses BC highways.

Why it matters

ICBC recorded 12,519 wildlife-vehicle collisions in BC in 2024 — a record, with 24 fatal. At 100 km/h a car needs 71 metres to stop on dry asphalt. The device must see, identify, and warn faster than a human could react and brake inside that distance. That single requirement drives every design choice: speed and reliability come before perfect accuracy.

How it works, step by step

1. See	The windshield camera streams live video of the road ahead.
2. Detect	A small AI model (YOLOv8n) running on a dedicated Hailo chip finds and labels animals in each frame, in real time.
3. Measure	The system estimates distance to the animal from how large it appears, using the known real-world height of each species.
4. Warn	Distance sets the alert level. The closer the animal, the louder and faster the beep and light.
5. Log	Every sighting is written to the SD card with GPS coordinates, time, species, and confidence.

Build order: every feature retained, sequenced by risk

No feature is dropped. They are ordered so the core safety loop — detect wildlife, warn the driver, log the encounter — works first, and the highest-risk additions come last. This protects the timeline: if time runs short, what slips is the riskiest extra, never the core.

Priority	Feature	Note
P1	Wildlife species detection (7 classes)	The heart of the project and the research contribution.
P1	Real-time inference on Hailo-8L	Must beat human reaction time; already trained at 82% mAP50.

P1	Distance via apparent size (thin-lens)	Reliable baseline; needs camera calibration + species heights.
P1	Three-tier alert (buzzer, LED, OLED)	The driver-facing output; audio is the primary signal.
P1	GPS-tagged SD logging (JSONL)	Produces the BC highway wildlife encounter dataset.
P1	Road test + validation metrics	What the project is graded on; mAP, false pos/neg rates.
P2	Time-to-contact (TTC) distance + fusion	Sharper distance under motion; fused conservatively with thin-lens.
P2	Adult vs child pedestrian sub-class	Head-to-body ratio on top of person detection.
P2	Vehicle-type classification	Comes largely from the COCO base model; low added cost.
P2	GPS speed-scaling of zone thresholds	Widen the warning zones as vehicle speed rises.
P3	FLIR Lepton thermal camera + fusion	Best night / low-visibility gain, but a whole extra sensor: built last.

Roboflow is not a dropped feature — it was a dataset tool, now replaced by the local build pipeline that produced the trained model. Same result, one less dependency.

Hardware

Component	Role
Raspberry Pi 5	Main computer
Hailo-8L AI Kit (13 TOPS)	Runs the AI model fast
Pi Camera Module 3 NoIR	Road-facing camera, sees at night
u-blox NEO-M8N GPS	Location, speed, time (still to acquire)
Buzzer + WS2812B LED + OLED	Driver warnings
12V-to-5V regulator + 128GB SD	Car power and storage

The three-tier warning

Zone	Distance	What the driver gets
1	over 80 m	Silent — logged to SD only
2	40–80 m	Amber pulsing light + two short beeps
3	under 40 m	Red flashing light + rapid beep + species on screen

Two safety rules: the system only warns for animals moving **toward** the car (a shrinking bounding box means it is leaving), and when several animals are in view it warns for the **highest-risk** one only. Everything seen is still logged.

Timeline to mid-August

Week	Dates	Goal
1	Jul 2–8	Order parts, flash Pi, verify Hailo, bench test peripherals
2	Jul 9–15	Quantize model to INT8, first custom inference on-device
3	Jul 16–22	Integrate distance, zones, alerts, GPS logging — end-to-end demo
4	Jul 23–29	Tune, harden, build the in-car rig
5	Jul 30–Aug 5	Highway 97 road test, day and night, collect GPS data
6–7	Aug 6–17	Analyse results, write report, buffer

Where the project stands

Done: dataset of 2,860 filtered images across 7 species; YOLOv8n trained (mAP50 0.823); full pipeline and documentation; repository cleaned to engineering standards.

Left: assemble hardware; quantize to INT8; write the detection, alert, and logging software; acquire the GPS module; run the road test; write the report.

Computer Engineering (Deepansh) — software, model, deployment, logging | Electrical Engineering (Aditya) — camera, GPS, power, calibration